

Round 4:

해킹 THE MISSING RESEARCHER

CTF Type: Digital Forensics

Difficulty: Medium

Estimated Time:

Participants: teams

Flag Format: FLAG{ . . . }

1. CTF INTRODUCTION

Challenge Name

The Missing Researcher

Story

Dr. Alex Thomas, a cybersecurity researcher, disappeared while working on an important research project.

Before disappearing, Alex uploaded several files to a shared research folder.

Investigators recovered the files, but something doesn't seem right.

The files appear to be ordinary documents, photographs, and research notes.

However, Alex was known for hiding messages inside innocent-looking files.

Your mission:

Investigate the evidence, follow Alex's trail, and recover the final flag.

2. FILES PROVIDED TO PARTICIPANTS

Give participants this folder:

```
missing_researcher/
```

```
|  
├── photo.jpg  
├── meeting.png  
├── report.docx  
├── notes.pdf  
└── evidence.png
```

Introduction

Contents:

THE MISSING RESEARCHER

Dr. Alex Thomas has disappeared.

The following files were recovered from his workstation:

photo.jpg
meeting.png
report.docx
notes.pdf
evidence.png

Alex left behind the following message:

"Don't trust what you see.
The truth is hidden inside the files."

Find the flag.

Flag format:
FLAG{...}

4. STAGE 1 — PHOTO.JPG

Objective

Participants need to inspect the metadata of the image.

The image itself should look completely ordinary.

Add the following metadata to the image:

Artist:
Alex Thomas

Description:
Look at the other picture.

Title:
6D 65 65 74 69 6E 67 2E 70 6E 67

A participant using:
exiftool photo.jpg

will discover the hexadecimal string.

They should recognize:
6D 65 65 74 69 6E 67 2E 70 6E 67

as hexadecimal to ASCII.(decoder1.py)

Decoded:

meeting.png

Hint 1

If participants are stuck:

"Pictures contain more information than just pixels."

Hint 2 (strong hint)

"Check the metadata."

5. STAGE 2 — MEETING.PNG

Participants now investigate:

meeting.png

The image should appear completely normal.

Hide the following message inside the image:

VGhlIHBhc3N3b3JkIGlzOiBkb2N1bWVudA==

This is Base64.

Decoded:

The password is: document

To keep the CTF beginner-friendly, provide:

extract.py

The script extracts the hidden message.

Running:

```
python3 extract.py meeting.png
```

produces:

Hidden message detected!

VGhlIHBhc3N3b3JkIGlzOiBkb2N1bWVudA==

Base24 to text msg (decoder2.sh)

Participants then decode it:

```
echo "VGhlIHBhc3N3b3JkIGlzOiBkb2N1bWVudA==" | base64 -d
```

Result:

The password is: document

Hint

"The image contains something that your eyes cannot see."

6. STAGE 3 — REPORT.DOCX

Participants open:

`report.docx`

The document should look like a completely normal cybersecurity report.

Example:

CYBERSECURITY INCIDENT REPORT

Researcher: Alex Thomas

The investigation began on Monday.

Several authentication systems were tested.

Access controls were reviewed.

Encryption mechanisms were analyzed.

Logging systems were inspected.

No obvious compromise was detected.

CONCLUSION

The system appears to be secure.

However, the DOCX contains an additional file:

`customXml/item1.xml`

Participants can discover this with:

```
unzip -l report.docx
```

They then extract it:

```
unzip report.docx -d report_extracted
```

Inside:

report_extracted/customXml/item1.xml

Contents:

```
<message>
The next clue is inside the PDF.
</message>
```

Therefore:

The next clue is inside the PDF.

Hint

"A DOCX file isn't really just a document."

Stronger Hint

"What happens when you unzip it?"

7. STAGE 4 — NOTES.PDF

Participants open:

notes.pdf

The PDF contains:

RESEARCH NOTES

```
01 - Access Control
02 - Encryption
03 - Logging
04 - Monitoring
05 - Incident Response
```

All systems were tested successfully.

Nothing unusual was detected.

However, there is hidden text in the PDF:

KEY: 435446

The text should be white on a white background so that it isn't immediately visible.

Participants can discover it using PDF text extraction or by selecting/copying all text.

The hexadecimal value:

435446

decodes to:

CTF

Therefore, the clue is:

CTF

This tells participants that the final challenge is associated with the last evidence file.

Hint

"Sometimes text can exist without being visible."

Stronger Hint

"Select all the text in the PDF."

8. STAGE 5 — EVIDENCE.PNG

The final image should look like a screenshot recovered from Alex's computer.

It can contain:

```
Research/  
Documents/  
Photos/  
Evidence/  
final.txt
```

Somewhere inside the image is a QR code.

The QR code contains:

```
FLAG{alex_left_more_than_a_message}
```

Participants scan the QR code.

They obtain:

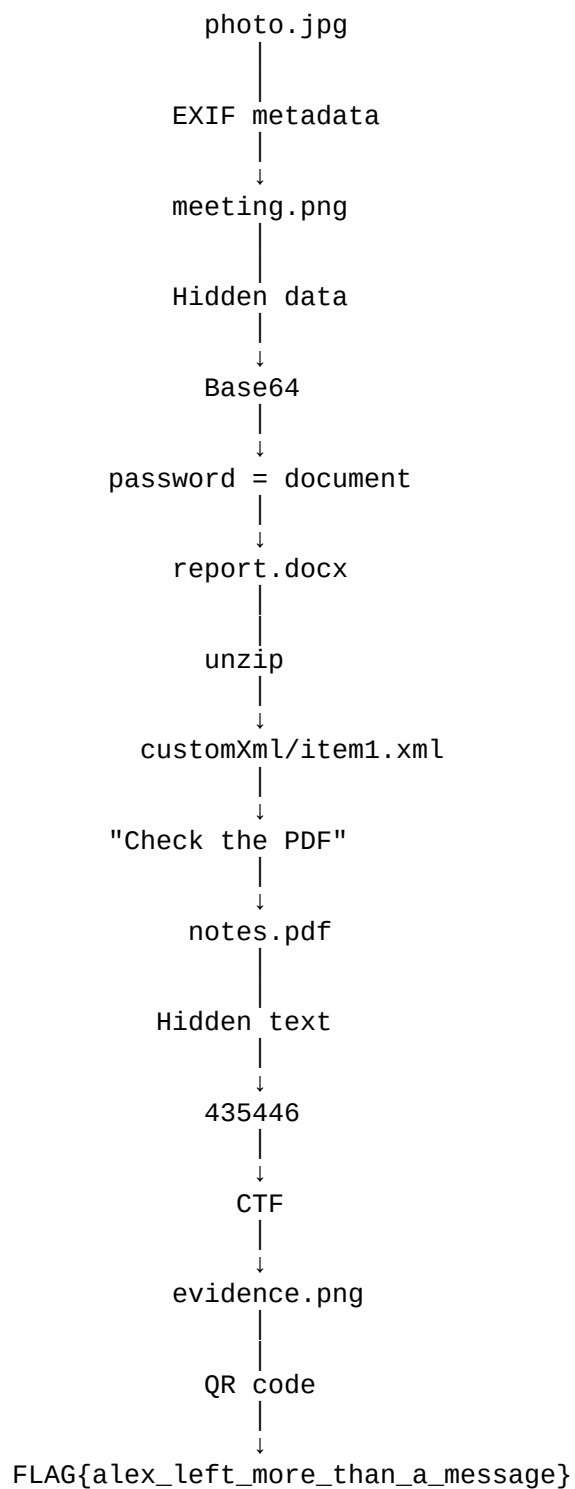
```
FLAG{alex_left_more_than_a_message}
```

讚 FINAL FLAG

```
FLAG{alex_left_more_than_a_message}
```

9. COMPLETE SOLVING PATH

The intended path is:



10. HINT SYSTEM

Don't immediately give players the answer.

Use three levels of hints.

Stage 1

Hint 1:

"Have you checked the properties of the image?"

Hint 2:

"Try exiftool."

Stage 2

Hint 1:

"There is data hidden in the image."

Hint 2:

"Look at the suspicious-looking text."

Stage 3

Hint 1:

"What type of file is DOCX?"

Hint 2:

"Try treating it as a ZIP archive."

Stage 4

Hint 1:

"The PDF may contain more text than you can see."

Hint 2:

"Try selecting everything."

Stage 5

Hint 1:

"Look carefully at the image."

Hint 2:

"Some things are easier for a camera to read than your eyes."

11. SCORING

Recommended scoring:

Stage	Points
Find EXIF clue	100
Decode Base64	100
Discover DOCX hidden file	200
Discover hidden PDF text	200
Find QR code	100
Final flag	300
Total	1000

12. ORGANIZER NOTES

The challenge intentionally teaches:

```
File Metadata
  ↓
Encoding
  ↓
Steganography
  ↓
Document Structure
  ↓
Hidden PDF Data
  ↓
Image Investigation
```

The important design principle is:

Every file should provide a clue that points toward the next file.

This prevents participants from randomly running tools against every file.

The challenge should feel like an investigation rather than a collection of unrelated puzzles.

13. TOOLS REQUIRED

Participants only need basic tools:

```
file
strings
grep
```

```
unzip  
exiftool  
python3
```

A QR scanner can be used for the final image.

14. FLAG SUBMISSION

At the end, participants submit:

```
FLAG{alex_left_more_than_a_message}
```

The first participant/team to submit the correct flag wins.